

IQR

2025 March

Week 10 Sunday 9

a) The interquartile range. compare the two interquartile ranges.

b) Any outliers in either set?

$$Q_3 - Q_1 = 82.5 - 56$$

$$\therefore \boxed{IQR = 26.5}$$

Potential Outliers:-

[Day class]

Lesser Outlier:-

$$Q_1 - 1.5 * IQR$$

$$= 56 - 1.5 \times 26.5$$

$$= \underline{\underline{16.25}}$$

Greater Outlier:-

$$Q_3 + 1.5 * IQR$$

$$= 82.5 + 1.5 \times 26.5$$

$$= \underline{\underline{122.25}}$$

$$\boxed{16.25, 122.25}$$

Here, in the given dataset, both Minimum (32) &

Maximum value (99) falls within the range [16.25, 122.25]

Therefore, no outliers in Day class.

[Night class]

Lesser Outlier:-

$$Q_1 - 1.5 * IQR$$

$$= 61.5$$

$$\boxed{61.5, 105.5}$$

Greater Outlier:-

$$Q_3 + 1.5 * IQR$$

$$= 105.5$$

Here, Minimum value (25.5) is not within this range, but, Maximum (98) falls with the range. Hence, 25.5 is an Outlier.

July : 25

M	T	W	T	F	S	S
1	2	3	4	5	6	
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

August : 25

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

September : 25

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

October : 25

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

November : 25

M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

December : 25

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				